Data Mining - MSA 2019

Assessment 1, Form: $\boxed{\text{A}}$

Name: ______________________________

Cohort: ______________________________

Date: ______________________________

Section 1. Multiple Choice (10 pts each)

1. When we are trying to predict a class target (either binary or otherwise) we generally...
   - (a) Determine the model parameters using training data and then evaluate the model's performance (accuracy) on all available data.
   - (b) Determine the model parameters using validation data and then evaluate the model's performance (accuracy) on test data.
   - (c) Determine the model parameters using all available data and then evaluate the model's performance (accuracy) on validation data.
   - (d) Determine the model parameters using training data and then evaluate the model's performance (accuracy) on validation data.
2. Which of the following is a drawback to leave-one-out cross-validation?
   - (a) Leave-one-out cross-validation doesn't give us a good sense for the error of the model on out-of-sample data.
   - (b) Leave-one-out cross-validation requires building the model and estimating parameters n times, where n is the number of observations in your data set.
   - (c) Leave-one-out cross-validation has been shown to always create models that suffer from high variance.
   - (d) Leave-one-out cross-validation only shows you how your model performed on one test observation, making it hard to draw conclusions about the generalizability of your model.
3. Suppose we find an association rule, "Milk => Bread" that has a lift statistic of 4. Which of the following interpretations of this statistic are correct?
   - (a) 4% of the people who bought Milk in my dataset also bought bread
   - (b) 4% of the people in my data bought Milk and then later bought bread
   - (c) 4% of the people in my data bought Milk and Bread
   - (d) Based on my sample, people are 4 times more likely to buy bread, given that they are buying milk, than they are to buy bread overall.
4. What does it mean to *prune* a Decision Tree?
   - (a) To grow the tree from the root node until the tree contains only one observation per leaf.
   - (b) To use a tree model for prediction of a categorical target.
   - (c) To remove nodes of the tree from the top so that we split the decision tree into several distinct trees.
   - (d) To remove nodes from the bottom of the tree to simplify the model and prevent overfitting of the training data.

5. What is the motivation behind using a Bonferroni adjustment to the p-values when creating a Decision Tree using the CHAID algorithm?

   (a) We can't trust the p-values because we don't have enough observations in our training data.

   (b) We can't trust the p-values because Chi-Square tests are known to be biased in the case of a binary response.

   (c) We're correcting for the family-wise error rate of running numerous chi-square tests and picking the most significant one.

   (d) We're correcting for the family-wise error rate of running several t-tests to find an ideal stopping place in the number of leaves to grow the tree.

6. Which of the following are reasonable strategies for missing value imputation for a predictive model? ***(Select all that apply)***

   (a) Purchase amount, a continuous variable, is missing in 3% of transactions. You fill in those missing values with the average purchase amount for all customers.

   (b) Income, a continuous variable, is missing in 15% of your observations. You have several outliers in this column, individuals with very large incomes. You fill in the missing values with the median income for all customers.

   (c) You've downloaded weather data that contains one observation for each month of the year for 20 years. One column, which contains the minimum temperature for each month is missing in 5% of cases. You fill in these missing values with the average of the minimum temperature column.

   (d) Insurance status, a binary variable indicating whether or not a customer has insurance, is missing for 10% of observations in your data. Since the majority of the customers in your table *do* have insurance, you simply assume this fact for the missing 10%.

Section 2. Short Answer

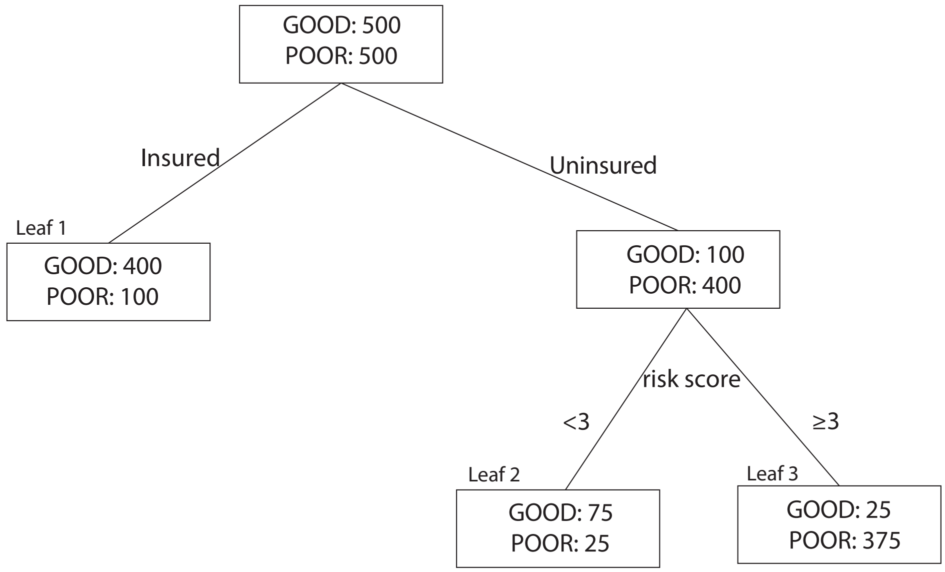
7. (15 pts.) Suppose 4% of the people who come in to my clinic have cuts and 5% have infections. Suppose 1% have both.

   a. What proportion _________ of the people coming into my clinic have neither cuts nor infections?

   b. What is the confidence _________ of the rule CUT => INFECTION? (i.e. people with cuts also have infections).

   c. What is the lift _________ of that rule?

8. (15 pts.) The following tree model seeks to model the probability of health care outcomes ('Good' vs. 'Poor') as a function of a patients insurance status and risk score. Use this tree to answer the questions that follow, if possible.



   a. What is the predicted probability that an insured individual with a risk score of 5 will have a good outcome?

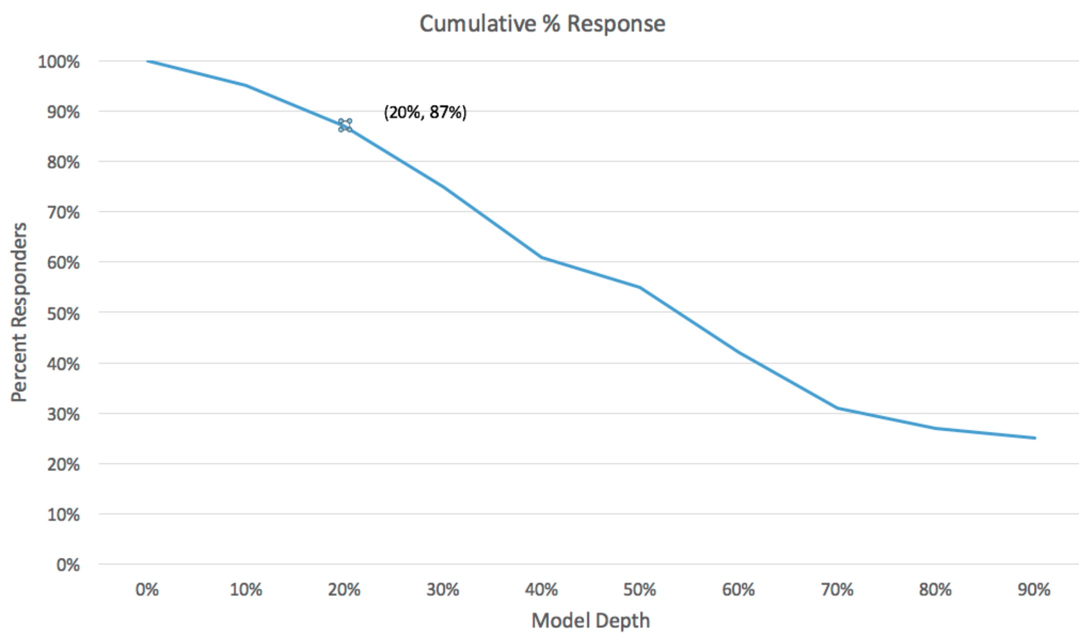
   b. Assuming a default cutoff probability of 0.5, what is the misclassification rate of this decision tree model?

   c. What is the odds ratio of insured vs. uninsured for having a good outcome? Present the odds ratio as follows: An individual with insurance has ______ times the odds of a good outcome compared to that of an uninsured person.

9. (10 pts.) Suppose you build a k-Nearest Neighbor model using two continuous input variables, x and y. The objective is to predict a binary target taking values of either *'Yes'* or *'No'*. For the following training data, compute the output decision for a test observation $\mathbf{x}_{test} = (5, 3)$ using the Manhattan distance function (1-norm), and 1 and 3 nearest neighbors. To make a prediction, use a simple voting majority.

(x,y)	**Target**
(1,6)	Yes
(2,4)	Yes
(3,7)	No
(6,8)	No
(7,4)	No
(8,5)	Yes

10. (10 pts.) Interpret the highlighted point on the following chart in a sentence using the given x and y coordinate as shown.



Section 3. True/False (10 pts each)

Mark each question as True or False.

_______ The choice for the hyperparameter k in a k-nearest neighbor model is important because if it is too small, your classification boundary will have high variance and the resulting predictions will be sensitive to noise and if it is too large than your classification boundary will have high bias and miss some of the pattern in the training data.

_______ One of the advantages of the k-nearest neighbor model is that it will provide us with a list of variables that are most important in predicting an outcome.

_______ Decision trees are not well suited to handle continuous input variables because it is not possible to split a node into two branches according to a continuous variable.

_______ For an ordinal variable such as a likert scale response with L different levels (and no missing values), Enterprise Miner will evaluate $L - 1$ potential splitting rules to determine the best split.

_______ One of the advantages of a Decision Tree is that it has the ability to use observations that have missing values on variables that are in the model.

_______ For an association rule like "Peanut Butter => Jelly", suppose that, in my dataset, 45% of the people who bought Peanut Butter also bought Jelly. This statistic represents the *support* of the association rule.

_______ If a leaf of a decision tree for a binary target contains 50% positive outcomes and 50% negative outcomes then this leaf has the highest possible value of entropy, gini, and misclassification rate.